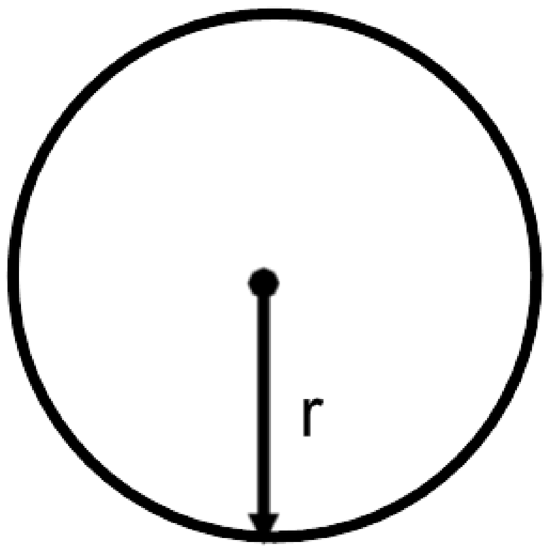
**Problem 1**

# Problem 1

**Problem 1** *The radius of a circle is increasing at a rate of 2 inches per minute.*




(a) *At what rate is the circumference of the circle changing when the radius is 10 inches?*

**Explanation.** We know: $\frac{dr}{dt} = 2$ inches per minute and we want to find $\frac{dc}{dt}$ when $r = 10$.

$$
\begin{aligned}
c &= 2\pi r \\
\frac{dc}{dt} &= 2\pi \frac{dr}{dt} \\
\left[\frac{dc}{dt}\right]_{r=10} &= 2\pi(2) \\
\left[\frac{dc}{dt}\right]_{r=10} &= 4\pi \text{ inches per minute}
\end{aligned}
$$

(b) *At what rate is the area of the circle changing when the radius is 12 inches?*

**Explanation.** We know: $\frac{dr}{dt} = 2$ inches per minute and we want to find $\frac{dA}{dt}$ when $r = 12$.

$$
\begin{aligned}
A &= \pi r^2 \\
\frac{dA}{dt} &= 2\pi r \frac{dr}{dt} \\
\left[\frac{dA}{dt}\right]_{r=12} &= 2\pi(12)(2) \\
\left[\frac{dA}{dt}\right]_{r=12} &= 48\pi \text{ inches squared per minute}
\end{aligned}
$$